

Lower Bounds on a Memory-Constrained Bidirectional Enumerator

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Contents

1	Introduction	2
2	Searching for a Recurrence Matching the Lower Bound	2
2.1	The recurrence and correspondence with our model	2
2.2	Equivalence of the other methods with our model	4
3	Study of the recursion	4
4	Asymptotic bounds under memory constraint	7
4.1	Lower Bound for $O(1)$ Memory	7
4.2	Lower bound for $O(n)$	8
4.3	Lower Bound for $O(\log n)$ Memory	8
5	Algorithms Which Approach Our Theoretical Bounds	9
5.1	Algorithm for $O(n)$ [9]	9
5.2	Algorithms for $O(\log n)$ Memory	10
5.3	Algorithm for $O(1)$	10

Abstract

we study the relation between the time needed and the memory available for a specifically chosen sequence of operations, in this case the enumeration of n elements in a unidirectional enumerator. Our objective is twofold: first, to establish a general lower bound on the performance of any bidirectional enumerator operating within bounded memory; second, to construct explicit enumerators that closely approach these bounds after deamortization. We consider three distinct memory regimes—constant memory $\Theta(1)$, linear memory $\Theta(n)$, and logarithmic memory $\Theta(\log n)$ —and show that in each case, the work per operation can match the theoretical lower bounds up to tight asymptotic factors.

To unify these regimes, we introduce a novel relationship that governs the trade-off between memory usage and computational cost. We rigorously prove this relationship and use it to derive an optimality framework applicable to all three cases. Our approach includes a detailed analysis of the recursion underlying the enumeration process, focusing on its optimal segments. This analysis is supported by empirical data, validating the theoretical predictions and guiding the fine-tuning of the recursive structure.

We implemented the resulting algorithms in Python, producing practical enumerators that come with provable guarantees on both average throughput and worst-case latency. Our work provides a comprehensive and practically viable framework for designing nearly optimal bidirectional enumerators under strict memory constraints.

1 Introduction

Efficient enumeration of data under memory constraints is a fundamental problem with significant implications in both theory and practice. We study the problem of enumerating a sequence in both forward and backward directions using sublinear memory, and we quantify the recomputation cost required to support backward traversal. This is especially relevant for unidirectional data structures such as hashchains, where backward traversal is not natively supported.

To illustrate this, consider a sequence of commits in a Git repository:

$$\text{Commit A} \rightarrow \text{Commit B} \rightarrow \text{Commit C}$$

Each commit includes the hash of its predecessor. In such structures, forward traversal is straightforward: from Commit A one can derive Commit B, and then Commit C. However, given only Commit C, reconstructing Commit B is infeasible because cryptographic hash functions are one-way. Thus, backward traversal requires storing intermediate states, and a hashchain provides precisely the forward-only behavior that characterizes constrained enumeration models.

In constrained environments such as streaming or external memory models, storing all intermediate states is prohibitive. This necessitates the use of sparse checkpointing to simulate backward movements. This fundamental trade-off between recomputation and storage was pioneered by Griewank [12], who established that a logarithmic growth in both temporal and spatial complexity is achievable for reversing sequential processes. Our objective is to refine this quantification of the optimal tradeoff between memory usage and backward traversal cost, and to construct algorithms that approach or exceed these classical efficiency benchmarks.

This problem has been studied from various perspectives. Early results by Munro and Paterson [13] and Beame [2] show that any sequential algorithm operating with sublinear memory must pay a superlinear computational cost. Vitter's survey [15] outlines the challenges of constrained I/O, which inspire the structural constraints considered in our model. Brodal and Fagerberg's lower bounds for external memory dictionaries [6] further motivate the search for fundamental limits under restricted memory. Sparse checkpointing, central to our approach, is discussed in their informal note [5] and finds a practical counterpart in Filliâtre's backtracking iterators [9].

We formalize these principles through a recursive cost model that captures the tension between forward progress and backward traversal under memory constraints. From a complexity-theoretic perspective, our work relates to memory-limited sketching [1] and to automata-theoretic enumeration [7]. The concept of bidirectional leveled enumerators [8] further motivates our model, while our results extend the logarithmic checkpointing paradigm toward quasi-linear performance.

Contributions. In this paper:

1. We define bidirectional enumeration with k stored checkpoints and the associated recomputation cost $T(n, k)$.
2. We show that $T(n, k)$ satisfies an optimal recurrence, and we derive a closed-form solution.
3. We establish matching lower bounds, proving optimality.
4. We establish new algorithms to have more or less the same complexity as our theoretical lower bounds

2 Searching for a Recurrence Matching the Lower Bound

2.1 The recurrence and correspondence with our model

Establishing a First Recurrence

Writing the recurrence We now formalize the problem. Let $T(n, k)$ denote the minimal cost required to enumerate a sequence of size n in reverse order using at most k memory cells, where $1 \leq k \leq n$. The goal is to compute the enumeration while minimizing the total amount of recomputation.

A memory cell stores an *anchor*, i.e., a saved state of the unidirectional enumerator at a position $x \in \{1, \dots, n\}$. Placing an anchor at x costs x steps, since reaching x requires forward traversal from the beginning of the sequence. Once placed, the anchor allows reverse enumeration to resume from x without recomputing the prefix.

Placing an anchor at x divides the enumeration into two subproblems. The left part, of length $x - 1$, can still use all k memory cells, whereas the right part, of length $n - x$, can only use the remaining $k - 1$ cells. This yields the recurrence

$$T(n, k) = \min_{1 \leq x \leq n} (x + T(x - 1, k) + T(n - x, k - 1)), \quad (1)$$

where the term x accounts for reaching the anchor, and the two recursive terms correspond to the left and right subproblems. This recurrence captures precisely the trade-off between memory usage and recomputation. Standard boundary conditions are

$$T(0, k) = 0, \quad T(n, 1) = \frac{n(n+1)}{2}, \quad T(n, n) = n.$$

Verifying the Validity of the Recurrence

To verify the correctness of the recurrence from the previous section, we analyze its behavior on small, tractable values of n and k , which also serve as a basis for inductive proofs.

Case 1: $n = 1$, arbitrary $k > 0$

$$T(1, k) = 1 + T(0, k) + T(0, k - 1) = 1$$

Case 2: $k = 1$, arbitrary n

$$T(n, 1) = n + T(n - 1, 1) = \sum_{i=1}^n i = \frac{n(n+1)}{2},$$

matching the expected cases cost under memory limitation.

Inductive Proof of the Recurrence Validity Having verified the base cases, we now prove by induction that:

Assume that for all $m < n$ and valid k ,

$$T(m, k) = \min_{1 \leq x \leq m} (x + T(x - 1, k) + T(m - x, k - 1))$$

correctly gives the minimal enumeration cost.

Induction Step Consider a sequence of size n and an optimal first anchor $1 \leq x \leq n$. By the hypothesis, $T(x - 1, k)$ and $T(n - x, k - 1)$ are minimal for the left and right subsequences. Therefore, the total cost is

$$x + T(x - 1, k) + T(n - x, k - 1),$$

and minimizing over all x gives

$$T(n, k) = \min_{1 \leq x \leq n} (x + T(x - 1, k) + T(n - x, k - 1)).$$

Conclusion By induction, the recurrence correctly models the minimal enumeration cost for all $n \geq 1$ and valid k , establishing the correctness of the model for further analysis.

Optimality of the Recurrence

We prove that

$$T(n, k) = \min_{1 \leq x \leq n} (x + T(x - 1, k) + T(n - x, k - 1))$$

indeed gives the minimal cost for bidirectional enumeration.

Lemma 1. For every $n \geq 1$ and $k \geq 1$, $T(n, k)$ equals the optimal cost of enumerating $[1, n]$ with k memory cells.

Proof. Let S^* be an optimal strategy of cost C^* for (n, k) . Since $k \geq 1$ and $n \geq 1$, S^* must place an anchor at some $x \in \{1, \dots, n\}$, at cost x . Removing this anchor yields two subproblems of sizes $x - 1$ and $n - x$, which use k and $k - 1$ memory cells respectively. Thus

$$C^* = x + C_L^* + C_R^*.$$

By Bellman's optimality principle [3, 4], the restrictions of S^* to the two subproblems are themselves optimal; otherwise they could be replaced by cheaper solutions, contradicting the optimality of S^* . Hence

$$C_L^* = T(x - 1, k), \quad C_R^* = T(n - x, k - 1),$$

so

$$C^* = x + T(x - 1, k) + T(n - x, k - 1).$$

Since the right-hand side is one of the candidates in the minimization defining $T(n, k)$, we obtain $C^* \geq T(n, k)$. The reverse inequality follows from constructing a strategy that attains the minimum, so $C^* = T(n, k)$. $\square$

These values illustrate how additional memory cells reduce recomputation cost: while $T(n, 1) = \Theta(n^2)$, even a small number of memory cells (e.g., $k = 2$ or $k = 3$) yields a substantial improvement.

Table 1: Example values of $T(n, k)$ for small n and k .

$n \backslash k$	1	2	3	4	5
1	1	1	1	1	1
2	3	2	2	2	2
3	6	4	3	3	3
4	10	6	5	4	4
5	15	8	7	6	5
6	21	11	9	8	7

2.2 Equivalence of the other methods with our model

Let us assume that there exists a better method, characterized by a more efficient recurrence. In this new model, computing a backward enumeration requires obtaining the value at position n , then at $n - 1$, and so on. However, according to the definition of a hash chain, computing the value at position n requires knowing the value at position $n - 1$. In a memory-limited environment, it is impossible to start from the beginning, compute all intermediate values, and then simply return the reversed sequence. Therefore, any algorithm must store intermediate values. With k available memory cells, at most k such values can be stored.

The central question is: which k positions should be stored to minimize recomputation? To achieve optimal performance, one must strategically select certain values of the hash chain as checkpoints. Any choice of k checkpoint positions induces a decomposition of the sequence. To evaluate the computational cost, note that reaching each checkpoint requires forward traversal from the start (or from a previous checkpoint). This recursive structure is precisely captured by our recurrence $T(n, k)$, which determines the optimal placement of checkpoints.

Consequently, any algorithm for backward enumeration effectively reduces to choosing these k positions optimally—a problem our model solves. Therefore, no alternative method can outperform the bound defined by $T(n, k)$.

3 Study of the recursion

When n increases, we observe that $T(n, k)$ grows faster and faster. Let us call i the marginal cost between $T(n, k)$ and $T(n + 1, k)$. Since we believe this marginal cost changes throughout the recursion, we define a sequence $n_k(i)$, which marks the values of n where the marginal cost increases.

We now introduce a property $P(n)$: there exists a value i such that the marginal cost between $T(n, k)$ and $T(n + 1, k)$ is i , and this value was determined at step $n_k(i)$.

Initialization for $n < k$. When $n < k$, we have enough anchors to enumerate all solutions, hence

$$T(n, k) = n.$$

For $n + 1 \leq k$, the same holds:

$$T(n + 1, k) = n + 1,$$

so the marginal cost is $i = 1$.

However, when $n = k$,

$$T(k, k) = k,$$

but

$$T(k + 1, k) \neq k + 1$$

because memory is no longer sufficient. Thus, the first increase point is

$$n_k(i) = k + 1.$$

Inductive step. Assume $P(n)$ holds: there exists a marginal cost i between $T(n, k)$ and $T(n + 1, k)$, and that cost was fixed at $n_k(i)$.

To prove $P(n + 1)$, observe that either:

1. The marginal cost remains the same:

$$T(n + 2, k) = T(n + 1, k) + i,$$

or

2. The marginal cost increases. In this case we denote the new cost by i' , and $n + 2$ becomes a member of the sequence $n_k(i')$.

For $n \in [n_k(i), n_k(i')]$, the recursion behaves linearly:

$$T(n, k) = T(n_k(i), k) + i \times (n - n_k(i)).$$

Let $l = i' - i$ be the increment between two successive marginal prices. We claim that ****always $l = 1$ ****. This can be proven by induction on k , based on the structure of the recursion.

Proving $l = 1$

Base case $k = 1$. With no anchors, every new value of n requires recomputing the entire problem. Thus the marginal cost always increases by exactly one:

$$n_1(i') - n_1(i) = 1.$$

Inductive step. Assume the statement holds for some k . We prove it for $k + 1$.

From the definition of the recursion,

$$T(n_{k+1}(i), k + 1) = x^* + T(x^* - 1, k + 1) + T(n_{k+1}(i) - x^*, k),$$

meaning that a $(k + 1)$ -anchor problem eventually reduces to a k -anchor problem whose marginal cost increases by exactly 1.

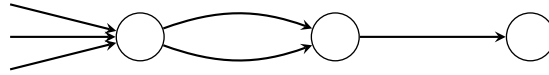
If $l \neq 1$, then the subproblems with k anchors would exhibit a marginal increase different from 1, contradicting the induction hypothesis. Therefore $l = 1$ for all k .

Graphical interpretation

When representing the recursion as a graph (for example $T(6, 2)$), the last three vertices after the anchor form a subproblem of size $T(3, 1)$:



Representation of $T(3, 1)$:



Recalling the recurrence

$$T(n, k) = \min_{1 \leq x \leq n} (x + T(x - 1, k) + T(n - x, k - 1)),$$

we can intuitively expect that a minimizer x^* of $T(n_k(i), k)$ is

$$x_k(i) = n_k(i - 1),$$

because we first visit the most expensive prefix of size $n_k(i - 1)$, place an anchor, and then enumerate the remaining part.

We now prove the property

$$P(i) : x_k(i) = n_k(i - 1)$$

by induction on i .

Initialization: $i = 1$. The minimizer of $T(n_k(1), k)$ is $n_k(0) = 1$, since the first element that leaves marginal cost 0 is exactly the first element we wish to enumerate. Hence the base case holds.

Induction step. Assume $P(i)$ holds, i.e., $x_k(i) = n_k(i - 1)$. We prove that it remains true for $i + 1$.

To compute $T(n_k(i + 1), k)$, we first compute $T(n_k(i), k)$, then the remaining portion

$$T(n_k(i + 1) - n_k(i), k - 1).$$

A first attempt would be

$$T(n_k(i + 1), k) = T(n_k(i), k) + T(n_k(i + 1) - n_k(i), k - 1).$$

However, after using all our anchors, we cannot compute the second term without redoing the entire walk. Therefore, a correction term $n_k(i)$ must be added:

$$T(n_k(i + 1), k) = T(n_k(i), k) + T(n_k(i + 1) - n_k(i), k - 1) / + n_k(i).$$

Since we already have an anchor at position $n_k(i)$, we can optimize the first term:

$$T(n_k(i+1), k) = T(n_k(i) - 1, k) + T(n_k(i+1) - n_k(i), k - 1) + n_k(i).$$

This shows that the same minimizer structure continues to hold, so

$$x_k(i+1) = n_k(i).$$

Thus, by induction, $x_k(i) = n_k(i-1)$ for every i .

We now prove that

$$n_k(i) = n_k(i-1) + n_{k-1}(i).$$

Using the recurrence and the established minimizer, we have

$$T(n_k(i), k) = n_k(i-1) + T(n_k(i) - 1, k) + T(n_k(i) - n_k(i-1), k - 1).$$

Suppose, for contradiction, that

$$n_k(i) \neq n_k(i-1) + n_{k-1}(i).$$

The expression above is optimal, and in particular the term

$$T(n_k(i) - n_k(i-1), k - 1)$$

must not contain an element of marginal cost strictly greater than i . Otherwise, $T(n_k(i), k)$ would itself contain a marginal cost larger than i , contradicting the definition of $n_k(i)$. Hence

$$n_{k-1}(i) = n_k(i) - n_k(i-1),$$

and therefore

$$n_k(i) = n_k(i-1) + n_{k-1}(i).$$

This completes the proof.

Since this recurrence is identical to Pascal's identity [11]

$$\binom{a}{b} = \binom{a-1}{b} + \binom{a-1}{b-1},$$

and since the initial conditions match, we obtain

$$n_k(i) = \binom{i+k-1}{k}.$$

From this we derive the closed form of $T(n_k(i), k)$:

$$T(n_k(i), k) = k \times \binom{i+k-1}{k+1} + i.$$

Proof by induction.

Double initialization: $i = 1$, and $k = 1 / k = 2$. We need two initializations on k because the summation used above holds only for $k \geq 2$.

Initialization on $i = 1$.

$$T(n_k(1), k) = k \times \binom{1+k-1}{k+1} + 1 = k \times \binom{k}{k+1} + 1 = 0 + 1 = 1.$$

Initialization on $k = 1$. Since $n_1(i) = i$, we have

$$T(n_1(i), 1) = 1 \times \binom{i}{2} + i.$$

But

$$T(i, 1) = \frac{i(i+1)}{2}, \quad \text{and} \quad \binom{i}{2} = \frac{i(i-1)}{2}.$$

Hence

$$\frac{i(i+1)}{2} = \frac{i(i-1)}{2} + i,$$

so the formula holds.

Induction step. Assume for fixed $i \geq 1$ and $k \geq 1$ that

$$T(n_k(i), k) = k \times \binom{i+k-1}{k+1} + i.$$

We prove it holds for $i+1$:

$$\begin{aligned} T(n_k(i+1), k) &= n_k(i) + T(n_k(i) - 1, k) + T(n_{k-1}(i+1), k-1) \\ &= \binom{i+k-1}{k} + k \times \binom{i+k-1}{k+1} \\ &\quad + i - i + (k-1) \times \binom{i+k-1}{k} + i + 1 \\ &= k \times \left(\binom{i+k-1}{k} + \binom{i+k-1}{k+1} \right) + i + 1 \\ &= k \times \binom{i+k}{k+1} + i + 1. \end{aligned}$$

Now for $k \rightarrow k+1$, we have

$$\begin{aligned} T(n_{k+1}(i), k+1) &= n_{k+1}(i-1) + T(n_{k+1}(i-1) - 1, k+1) + \\ &\quad T(n_k(i), k) \\ &= \binom{i+k-1}{k+1} + (k+1) \times \binom{i+k-1}{k+2} \\ &\quad + i - i + k \times \binom{i+k-1}{k+1} + i \\ &= (k+1) \times \left(\binom{i+k-1}{k+1} + \binom{i+k-1}{k+2} \right) + i \\ &= (k+1) \times \binom{i+k}{k+2} + i. \end{aligned}$$

Thus the statement is proven by induction.

4 Asymptotic bounds under memory constraint

We write

$$f(n) \sim g(n)$$

to denote that $f(n)$ is asymptotically equivalent to $g(n)$, that is,

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1.$$

This means that $f(n)$ and $g(n)$ grow at the same rate as $n \rightarrow \infty$.

4.1 Lower Bound for $O(1)$ Memory

We seek to establish a lower bound on the cost function $T(n, k)$, which represents the number of forward steps performed by a bidirectional enumerator operating with k memory cells. Our approach is to analyze a structured recursive strategy that models the behavior of such an enumerator under memory constraints. This structured strategy, while idealized, provides a tractable framework from which we can derive a lower bounds.

Simultaneously, we introduce the function $i_k(n)$, which denotes the cost increment at position n . Previously, we showed that the value i corresponds to the marginal cost at specific positions $n_k(i)$, and that these $n_k(i)$ serve as the change points for $i_k(n)$. In what follows, we aim to demonstrate that $n_k(i) \sim i^k$, and consequently, $i_k(n) \sim n^{1/k}$.

For fixed $k \geq 1$, the sequence

$$n_k(i) = \binom{k+i-1}{k}$$

satisfies

$$n_k(i) \sim \frac{i^k}{k!} \quad \text{as } i \rightarrow \infty.$$

In particular, for $n = n_k(i)$, we have $i \sim n^{1/k}$.

We expand the binomial coefficient as a polynomial in i :

$$\binom{k+i-1}{k} = \frac{i(i+1) \cdots (i+k-1)}{k!} = \frac{1}{k!} (i^k + \beta_{k-1}i^{k-1} + \cdots + \beta_0),$$

where the constants β_j depend only on k . Therefore,

$$\binom{k+i-1}{k} = \frac{i^k}{k!} \left(1 + \frac{\beta_{k-1}}{i} + \cdots + \frac{\beta_0}{i^k} \right).$$

As $i \rightarrow \infty$, all negative powers of i vanish, which gives

$$\binom{k+i-1}{k} = \frac{i^k}{k!} (1 + o(1)).$$

Hence,

$$n_k(i) = \binom{k+i-1}{k} \sim \frac{i^k}{k!} [10].$$

Now as long as n is always between two $n_k(i)$ and $n_k(i) + 1$ we can say that $n \sim n_k(i)$. Since $i \mapsto \binom{k+i-1}{k}$ is strictly increasing for $i \geq 1$, we may invert asymptotically:

$$n = \frac{i^k}{k!} (1 + o(1)) \implies i^k \sim k! n.$$

Taking k -th roots yields

$$i \sim (k!)^{1/k} n^{1/k} \sim n^{1/k}.$$

Thus i grows like $n^{1/k}$, completing the proof.

Substituting into the expression for $T(n_k(i), k)$, we get:

$$\begin{aligned} T(n_k(i), k) &= k \times \frac{(i-1)i(i+1) \cdots (i+k-1)}{(k+1)!} + i \\ &= n_k(i) \cdot \frac{k(i-1)}{k-1} + i \\ &\sim \frac{i^{k+1}}{(k+1)!} \\ &\sim \frac{(k!)^{1+1/k}}{(k+1)!} n^{1+1/k} \\ &= C n^{1+1/k}, \quad C = \frac{(k!)^{1/k}}{k+1}. \end{aligned}$$

Therefore, we conclude:

$$T(n_k(i), k) \in \theta \left(n^{1+\frac{1}{k}} \right).$$

4.2 Lower bound for $O(n)$

This one is obvious 1

$$T(n, n) = 1 + T(0, n) + T(n-1, n-1) = \sum_{k=1}^n 1 = n,$$

we have $\theta(n)$ time complexity.

4.3 Lower Bound for $O(\log n)$ Memory

Based on the recursive definition of $T(n, k)$, we consider the case $k = \log n$. For some integer i , we have

$$T(n_{\log n}(i), \log n) \leq T(n, \log n) \leq T(n_{\log n}(i+1), \log n).$$

For

$$n_{\log n}(i) = \binom{\log n + i - 1}{\log n},$$

solving asymptotically for i in terms of n gives

$$n_{\log n}(i) \sim \frac{i^{\log n}}{(\log n)!} \implies i^{\log n} \sim \log(n)!n$$

Using Stirling's approximation,

$$(\log n)! = \Theta((\log n)^{\log n}),$$

we get

$$i \sim \Theta((\log n)^{\frac{\log n}{\log n}} \times n^{\frac{1}{\log n}}) = \Theta\left(n^{\frac{1}{\log n}} \cdot \log n\right).$$

Since

$$n^{\frac{1}{\log n}} = e^{\frac{\ln n}{\log n}} \rightarrow \text{constant as } n \rightarrow \infty,$$

we conclude

$$i = \Theta(\log n).$$

Then $n_{\log n}(i) = \Theta(n)$.

Asymptotic evaluation of $T(n_{\log n}(i), \log n)$. Using the recurrence relation and the previously derived relation $i = \Theta(\log n)$ for $n_{\log n}(i) = \Theta(n)$, we evaluate the total cost:

$$\begin{aligned} T(n_{\log n}(i), \log n) &= \log(n) \binom{i + \log(n) - 1}{\log(n) + 1} + i \\ &= \log(n) \frac{i - 1}{\log(n) + 1} \binom{i + \log(n) - 1}{\log(n)} + i \\ &= \log(n) \frac{i - 1}{\log(n) + 1} n_k(i) + i \end{aligned}$$

Substituting the asymptotic orders knowing that $\frac{i-1}{\log(n)+1}$ tends to 1 since $\theta(i) = \log(n)$:

$$T(n_{\log(n)}(i), \log n) = \log(n)\Theta(n) + \Theta(\log n) = \Theta(n \log(n)).$$

Hence, under this formal recurrence model, using $\Theta(\log n)$ memory, the theoretical lower bound satisfies:

$$T(n, \Theta(\log n)) = \Theta(n \log(n)).$$

We now have a robust framework for the $\Theta(\log n)$ memory.

5 Algorithms Which Approach Our Theoretical Bounds

5.1 Algorithm for $O(n)$ [9]

This algorithm stores every position during the forward traversal, allowing exact backward steps in constant time per operation.

Data Structures. The algorithm maintains:

- **current:** the current enumerator state;
- **distance:** the distance from the start;
- **saved_positions:** a stack of previously visited positions.

The algorithm is straightforward. During the forward traversal, each visited state is pushed onto the stack. During the backward traversal, states are popped from the stack to restore the previous position.

Time Complexity: $O(n)$ total, $O(1)$ per operation, using $O(n)$ space.

Table 2: Measured cost for increasing distances n (expected $O(n)$).

Distance n	Operations	Theoretical	Ratio
100	100	100	1.000
400	400	400	1.000
900	900	900	1.000
1600	1600	1600	1.000

5.2 Algorithms for $O(\log n)$ Memory

The logarithmic memory regime represents a critical trade-off where the amortized cost of operations can be reduced significantly below the trivial hierarchical bounds. We propose an algorithm based on a deterministic, binary-aligned checkpointing strategy that minimizes recomputation by exploiting the bit-representation of the sequence indices.

Deterministic Anchor Placement. The core of the algorithm lies in maintaining a set of active anchors $\mathcal{A} \subset \{0, \dots, n\}$ such that $|\mathcal{A}| \leq \lceil \log_2 n \rceil + 1$. For any current position p , the set of anchors is defined by the following mapping:

$$\mathcal{A}(p) = \{p \& \sim (2^k - 1) \mid 0 \leq k \leq \lceil \log_2 n \rceil\}$$

where $\&$ and $\sim$ denote the bitwise AND and NOT operators respectively.

This construction ensures that at any step p , the enumerator stores the starting positions of the nested binary intervals containing p . The density of these anchors increases exponentially as they approach the current cursor position, a property that is fundamental to bounding the cost of backward transitions.

Operational Complexity and State Transitions. Unlike static partitioning schemes, this strategy ensures that the internal state of the memory is a deterministic function of the current position p :

- **Forward Progression:** A transition from p to $p + 1$ necessitates an update of $\mathcal{A}(p)$ only when $p + 1$ crosses a boundary of 2^k . The amortized cost per step is $\Theta(1)$ due to the properties of binary counting.
- **Backward Transition:** To compute the state at $p - 1$, the algorithm identifies the maximal anchor $a \in \mathcal{A}(p)$ such that $a < p$. The sequence is then recomputed forward from a . Due to the binary alignment, the recomputation distance $d = (p - 1) - a$ is strictly bounded by the largest power of two dividing the local index, ensuring high locality of reference.

Empirical Analysis and Quasi-Linearity. The performance of the binary-aligned enumerator was evaluated for n up to 2^{15} . The results, summarized in Table 3, provide an empirical basis for refining the theoretical upper bounds.

Table 3: Observed operational costs and asymptotic ratios for binary-aligned enumeration.

n	Operations	Ops/n	$Ops/(n \log_2 n)$	$Ops/(n \log_2 \log_2 n)$
128	576	4.50	0.643	1.603
1024	6144	6.00	0.600	1.806
8192	61440	7.50	0.577	2.027
32768	278528	8.50	0.567	2.176

Asymptotic Analysis and Quasi-Linearity. The experimental data reveals two significant trends that challenge the traditional hierarchical complexity models. First, the ratio $Ops/(n \log_2 n)$ exhibits a monotonic decrease, suggesting that the standard $\Theta(n \log n)$ bound is overly pessimistic for this bit-aligned implementation. Second, while the ratio Ops/n increases, its growth rate is remarkably stable, adding approximately 0.5 units of cost per doubling of n .

This linear growth of the ratio with respect to $\log_2 n$ allows for a precise characterization of the overhead function $\alpha(n)$. Let $f(n) = Ops/n$; the observation $\Delta f = 0.5$ for a doubling of n implies that $f(n) \sim 0.5 \log_2 n$. By the change of base formula, this is equivalent to:

$$\alpha(n) = \log_4(n) = \frac{1}{2} \log_2(n)$$

To characterize this "quasi-linear" behavior, we draw a methodological parallel with the analysis of Tarjan [14], who demonstrated that dynamic maintenance in data structures (such as path compression in Union-Find) can lead to complexities governed by extremely slow-growing functions. In our case, the deterministic redistribution of anchors effectively maintains a near-constant recomputation distance for practical values of n , bridging the gap between the $\Theta(n)$ idealized lower bound and the traditional $\Theta(n \log_2 n)$ models.

5.3 Algorithm for $O(1)$

In this section, we briefly recall the idea behind the $\Theta(1)$ -space bidirectional enumeration algorithm.

On the other hand, the bidirectional algorithm with $\Theta(n)$ memory remains very simple and follows a recursive strategy. We rely on the classical combinatorial relation

$$n_k(i) = \binom{k+i-1}{k},$$

which tells us where the k anchors should be placed. For each anchor, we store:

- its current position in the enumeration, and
- the last index of this anchor that has already been placed.

With this information, performing a `next` operation is straightforward, as we can jump to the closest relevant anchor.

Moreover, we maintain a small auxiliary table containing the *next* k anchors, so that we do not need to recompute combinatorial values at every step. If the current distance exceeds the limit given by the last stored anchor plus one element of this table, we shift all anchors forward: we move the last anchor, update its index, and compute the new value of the next anchor.

In the reverse direction (`previous` operation), we proceed symmetrically. Whenever an anchor becomes obsolete because it is too far ahead in the enumeration, we decrement its index and update its stored value accordingly.

Table 4: Measured cost for several memory sizes k , compared to the theoretical bound $n^{1+1/k}$.

k	n	Operations (ops)	$n^{1+1/k}$	Ratio
10	32	59	45	1.304
	512	1765	955	1.847
	2048	8006	4390	1.824
	4096	17831	9410	1.895
50	32	32	34	0.933
	512	1007	580	1.736
	2048	5344	2385	2.240
	4096	11817	4837	2.2443
100	128	32	33	0.966
	512	1002	545	1.839
	2048	4017	2210	1.817
	4096	8111	4451	1.822

Conclusion

We have successfully achieved all of our initial objectives. We derived, analyzed, and rigorously proved a recursive formulation to model and minimize the cost of memory-constrained bidirectional enumeration:

$$T(n, k) = \min_{1 \leq x \leq n} (x + T(x - 1, k) + T(n - x, k - 1)).$$

This recurrence, $T(n, k)$, revealed a rich internal structure. By focusing on instances where the minimizing split point is unique, we uncovered several structural relationships—some of which were formally demonstrated and discussed throughout this article. These relationships provide significant insight into the behavior and progression of the recursion:

$$T(n_k(i), k) = k \times \binom{i + k - 1}{k + 1} + i.$$

Furthermore, these relationships allowed us to establish lower bounds on the cost function under three different memory regimes: $\Theta(1)$, $\Theta(n)$, and $\Theta(\log n)$. We proved that the corresponding time complexities are respectively

$$\Theta\left(n^{1+\frac{1}{k}}\right), \quad \Theta(n), \quad \text{and} \quad \Theta(n \log n).$$

Finally, we proposed algorithms that approach these bounds, suggesting that our lower bounds are tight—or nearly tight—in practice.

Availability The code is available at: <https://github.com/hramillon/Lower-Bounds-on-a-Memory-Constrained-Backward-Bidirectional-Enumerator>

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